

Library Management System API

Project Overview

The goal of this project is to build a RESTful Library Management System using ASP.NET Core Web API, Entity Framework Core (Code First), SQL Server, and Clean Architecture.

The system should allow librarians to manage books, authors, categories, members, and borrowing operations through API endpoints. Authentication is optional and can be added as a bonus feature.

Functional Requirements

Book Management

- Add a new book
- Update book information
- Delete a book
- Retrieve all books
- Retrieve a single book
- Search books by title
- Search books by ISBN
- Filter books by category
- Filter books by author
- Check book availability

Author Management

- Add authors
- Edit authors
- Delete authors
- View all authors
- View author details
- List all books written by an author

Category Management

- Create categories
- Update categories
- Delete categories
- View categories
- Retrieve books belonging to a category

Member Management

- Register members
- Update member information

- Delete members
- View member profile
- List all borrowed books for a member

Borrowing Management

- Borrow a book
- Return a book
- Prevent borrowing unavailable books
- Calculate due date
- Track borrowing history

Authentication (Optional Bonus)

- Login
- JWT Authentication
- Authorization
- Roles: Admin and Librarian

Project Deliverables

- Working ASP.NET Core Web API
- SQL Server database created using Entity Framework Core migrations
- CRUD operations for all required modules
- Well-structured code
- GitHub repository with commits
- Swagger documentation (optional)
- Authentication and authorization (optional bonus)

Recommended Learning Resources

- Create your first ASP.NET Core Web API: <https://learn.microsoft.com/en-us/aspnet/core/tutorials/first-web-api>
- ASP.NET Core Documentation: <https://learn.microsoft.com/en-us/aspnet/core/>
- Entity Framework Core Documentation: <https://learn.microsoft.com/en-us/ef/core/>
- EF Core Migrations: <https://learn.microsoft.com/en-us/ef/core/managing-schemas/migrations/>
- Patrick God (YouTube): <https://www.youtube.com/@PatrickGod>
- Kudvenkat (YouTube): <https://www.youtube.com/@kudvenkat>
- Nick Chapsas (YouTube): <https://www.youtube.com/@nickchapsas>
- Milan Jovanović: <https://www.milanjovanovic.tech>